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Attending to emotional expressions: no evidence for automatic capture in the dot-probe task

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ABSTRACT

Research on automatic attention to emotional faces offers mixed results. Therefore we examined validity effects for facial expressions of different emotions (compared to neutral faces) with a dot-probe paradigm in seven studies (total $N = 308$). Systematic variations of type of emotion, CTI, task, cue size, and masking allow for a differentiated assessment of attentional capture by emotions and possible moderating factors. Results indicate a general absence of emotional validity effects as well as a lack of significant interactions with either of the manipulated factors, indicating that facial expressions of emotions do not capture attention in a fully automatic fashion. These findings suggest that situational and contextual factors have to be taken into account when investigating attentional capture of emotional faces.

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Automatic attention allocation to facial expressions of emotion has interested researchers from various fields of psychology for a number of reasons. First, one of the core functions of emotions is to convey information to interaction partners regarding behavioural dispositions, attitudes, and evaluations of the sender ("social communicative function of emotions": Chovil, 1997; Oatley, 1988). From this perspective, it seems natural to expect that perceivers are reciprocally tuned to automatically detect characteristic expressions of emotions in others. Attentional capture of emotional expressions would provide an advantage in the ability to detect this important information, which is helpful for predicting the behaviour of an interaction partner, being prepared for whatever this person might intend, and to respond quickly and adequately to his or her emotional state (Staugaard, 2010). Furthermore, researchers interested in identifying the cognitive bases of psychopathology also have proposed various hypotheses relating attentional biases towards specific expressions of emotions to individual differences with regard to psychopathological symptoms (Edalati, Walsh, & Kosson, 2016). Specifically, it was proposed that attentional capture by angry faces should relate to anxiety and social

phobia (see Bar-Haim, Lamy, Pergamin, Bakermans-Kranenburg, & van IJzendoorn, 2007; Liu, Qian, Zhou, & Wang, 2006; Rapee & Heimberg, 1997). In the present study, however, our focus is exclusively on general effects of attentional capture in non-clinical (normal) samples.

Several paradigms have been used to investigate attentional biases for facial expressions of emotion. The most prominent and widely used paradigms are (a) the visual search task (Treisman & Souther, 1985), (b) interference tasks like the flanker task (Eriksen & Eriksen, 1974) or the emotional Stroop task (Gotlib & McCann, 1984), and (c) different versions of the cueing task (Posner, 1980).

Variants of the search task that presented facial stimuli with different emotions have typically compared attentional capture for angry and happy faces. Findings are mixed, with some studies indicating an advantage for angry compared to neutral faces (Eastwood, Smilek, & Merikle, 2001; Fox et al., 2000; Hansen & Hansen, 1988; Horstmann & Bauland, 2006; Huang, Chang, & Chen, 2011; Öhman, Lundqvist, & Esteves, 2001), but sometimes also a reversed asymmetry (Juth, Lundqvist, Karlsson, & Öhman, 2005).¹ In addition to the heterogeneous findings, the search

literature is typically concerned with alternative explanations of their findings in terms of confounding perceptual features of the materials that are presented (Harms & Bundesen, 1983; Horstmann & Bauland, 2006; Miller, 1991; Purcell, Stewart, & Skov, 1996). Importantly, the search task has typically not been used to compare automatic attention allocation between emotional and neutral faces.

Research on automatic attention allocation to facial expressions of emotion with interference paradigms is rare. Fenske and Eastwood (2003) presented positive (happy), negative (angry), and neutral schematic faces in a flanker task and found stronger compatibility effects for positive than for negative flankers, that is, while positive flankers facilitated responding to positive targets but not to negative targets, negative flankers did not differentially influence responding to positive (incompatible) or negative (compatible) targets. This asymmetry, however, cannot be unambiguously attributed to the emotional content of the faces and seems to reflect effects of differences in perceptual features that are unrelated to emotions (Horstmann, Borgstedt, & Heumann, 2006). Corroborating this ambiguity, the finding of differential flanker compatibility effects for faces with positive vs. negative expressions was not replicated in a study by Moser, Huppert, Duval, and Simons (2008), who used photos of natural faces as stimuli and found parallel flanker compatibility effects for positive (reassuring) and negative (threatening) flanker faces.

In cueing tasks (Posner, 1980), participants have to classify a target that can occur at one of two locations. Before the appearance of the target, a cue is presented that either does (valid trial) or does not (invalid trial) indicate the position of the following target. The difference in responding to valid and invalid trials (validity effect) indicates automatic attention allocation. The most widely used cueing paradigm is the so-called dot-probe task that was originally developed by MacLeod, Mathews & Tata (1986). In the dot-probe task, two cues (one emotional and one neutral) are presented simultaneously. Trials in which the target appears at the spatial position of the emotional cue are defined as valid, whereas trials in which the target appears at the position of the neutral cue are defined as invalid. Participants either indicate the location of the target in a detection task or classify special features of the target, such as form or orientation, in a discrimination task. The dot-probe task is of particular interest to researchers investigating automatic attentional capture for emotional

stimuli, because it has some interesting features (cf. Fox, Derakshan, & Standage, 2012; Yiend, 2010): (a) It is suitable to measure automatic attention to emotional stimuli (due to the fact that emotional stimuli are completely task irrelevant), (b) it allows for an estimation of “pure” attentional effects that are free from behavioural interrupt effects (because the same emotional cues are presented in both, valid and invalid conditions, and thus main effects that are due to general alertness or interrupt do not enter into the validity effect), (c) it allows for an assessment of very fast acting attentional processes in the domain of automatic attention allocation that are free from strategic manipulation (if short cue-target intervals are chosen), and (d) in combination with appropriate baseline conditions (i.e. a neutral/neutral condition), the dot-probe paradigm also allows for a separate identification of engagement and disengagement components of attention.

Previous research with the dot-probe task using facial expressions of emotions as stimuli revealed a mixed picture: The dot-probe task assesses attentional capture of emotional stimuli via the so-called validity effects, that is, faster classifications of probes at the position of emotional cues compared to the position of neutral cues. Those attentional capture effects are an often reported finding in (socially) anxious populations (Arndt & Fujiwara, 2012; Bar-Haim et al., 2007; Liu et al., 2006; Torrence, Wylie, & Carlson, 2017; Zhao, Zhang, Chen, & Zhou, 2014). In healthy participants, however, effects are not that clear cut (Arndt & Fujiwara, 2012; Bar-Haim et al., 2007; Mogg, Philippot, & Bradley, 2004). Besides some few studies that reveal findings rather analogous to anxious populations (Bocanegra, Huijding, & Zeelenberg, 2012; Bradley, Mogg, Falla, & Hamilton (1998); Lacreuse, Schatz, Strazzullo, King, & Ready, 2013), there is a multitude of investigations that show no effects in normal participants (Bar-Haim et al., 2007; Kappenman, Farrens, Luck, & Proudfit, 2014; Koster, Verschuere, Burssens, Custers, & Crombez, 2007; Lien, Taylor, & Ruthruff, 2013; Mather & Carstensen, 2003).

The studies that examined the effect of emotional faces in the dot-probe task differ in several respects. One important factor is timing (cue-target interval, CTI, i.e. the time interval between the onset of the cue and the onset of the target). This variable is of particular importance because the rationale and interpretation of the resulting validity effect can switch with longer CTIs due to what is termed the “inhibition of return” effect in the literature (Posner

& Cohen, 1984). Whereas attention is allocated automatically to one of the two cues during the initial presentation, participants often try to move their attention back strategically to the central position if the presentation of the cues is extended for a longer duration (CTI > 300 ms). This removal of attention away from the initially attended cue has the paradoxical effect that re-allocation of attention to the initial position is delayed, sometimes resulting in a reversed validity effect for longer CTIs. Evidence for the crucial role of time was reported by Cooper and Langton (2006) who found a threat bias at 100 ms CTI but reported reversed effects for the 500 ms condition. Analogously, Stevens, Rist, and Gerlach (2009) demonstrated an angry-face-bias with a CTI of 175 ms but not 500 ms. Carlson and Reinke (2008) find effects in a non-anxious sample with a CTI of 133 ms and backward-masked cue presentation. In a study by Torrence et al. (2017) with 300 ms CTI, both happy and fearful faces captured attention. Mogg, Philippot, & Bradley (2004) find vigilance for angry faces with a CTI of 500 ms but not of 1250 ms. Studies with longer CTIs mostly report the absence of dot-probe effects (Gotlib et al., 2004; Pineles & Mineka, 2005). A special case of very short presentation time is a masked procedure, where not only the cues vanish rapidly but also the representation in iconic memory is overwritten by an irrelevant stimulus that masks the cues. Carlson and Reinke (2008) found attentional capture of emotional faces in a dot-probe experiment with 33 ms cue presentation and neutral faces as masks. Fox, Russo, and Dutton (2002) presented participants with masked happy, angry, and neutral faces in a single cueing paradigm and found a validity advantage for short CTIs (300 ms) that switched into an invalidity advantage due to inhibition of return for long CTIs (960 ms); the strengths of both these effects, however, did not differ between happy, angry, and neutral faces, indicating that the effect was not driven by the emotional content of the faces. Mogg and Bradley (1999, 2002) found validity effects of threatening faces with a CTI of 14 ms respectively 17 ms and jumbled faces as masks in socially anxious participants. As masking allows for measuring preconscious and automatic (in the sense of unconscious) processes, this technique was used in one of our experiments, too.

Most dot-probe studies examined the effect of angry or threatening faces (Bocanegra et al., 2012; Koster, Crombez, Verschuere, & De Houwer, 2004) but there are also some investigations with happy (Fox et al.,

2002; Neel, Becker, Neuberg, & Kenrick, 2012; Schultebrucks et al., 2016), sad (Schultebrucks et al., 2016), and fearful faces (Carlson & Reinke, 2008, 2014). From a theoretical point of view, the investigation of attentional capture for happy faces is also particularly interesting because happy faces are an important indicator of social affiliation, possible support and peaceful intentions. While some studies find a threat bias (see Bar-Haim et al., 2007), Bradley et al. (1997) find a bias away from threat. Analogously Cooper and Langton (2006) report a bias away from happy faces while Staugaard (2009) finds a happy face bias.

Our research originally attempted to establish a variant of the dot-probe task that produces robust and reliable effects of attentional capture for facial expressions of different emotions (compared to neutral faces). We wanted to use this paradigm for subsequent studies testing moderating factors of these effects for specific emotions; in particular, we were interested in whether filtering emotional face stimuli so that faces contain only information in low vs. high spatial frequency bands (see Goffaux & Rossion, 2006) has a systematic influence on automatic attentional capture for specific emotions (Smith & Schyns, 2009). Since our ultimate aim was to compare effects of attentional capture for different emotions, we included faces with angry, happy, and fearful expressions. We started out with what we thought was a standard variant of the dot-probe task. Based on frustrating initial findings, we then varied different features of the paradigm, ending up with a series of seven experiments, all of which failed to show any indication of attentional capture for any kind of validity effect for emotional expressions (both general and specific) in the dot-probe paradigm. Our aim with the current paper is to document the results of this series of experiments, and to discuss their implications for our understanding of automatic processing of emotional expressions and their potential moderating conditions (individual differences, contextual factors).

Method

All experiments reported in this paper had the same basic structure. In order to demonstrate that our findings account for a wide range of experimental conditions, we varied certain aspects between and within studies. All experiments were dot-probe studies with facial cues displaying different emotions. In the following we will explain the common features first and describe the differentiating dimensions

afterwards. Exact parameters for every experiment can be found in [Table 1](#).

Sample

Three hundred and eleven participants from different faculties were recruited at the campus building of the FSU Jena and received chocolate and money for their participation. Participants with error rates that were extreme values following Tukey's (1977) criterion (>3 interquartile ranges above the third quartile of the inter-individual distribution within each study) were discarded from the analysis. This entailed an exclusion of three participants (one participant in Experiments 1, 2, and 6, respectively). Additionally, 21 participants who did not manage to meet the criteria to proceed with the experiment during the practice trials ($<10\%$ errors; $RT < 700$ ms) in Experiments 5 and 6 were not allowed to complete the experiment and thus produced no data that could have entered into the analysis.

Materials

All stimuli were presented on a 17" CRT Monitor (XGA resolution 85 Hz) with the Psychopy software (Peirce, 2007, 2009). Frontal face photographs from the Radboud Faces Database (Langner, Dotsch, et al., 2010) were chosen as cue stimuli. Every experiment contained a set of male and female greyscale faces of 10 identities per gender (see [Appendix B](#) for an example and [Appendix C](#) for a list of all used identities). While faces were presented as rectangular photographs in Experiments 1, 2, 3, 6, and 7, oval

cutouts were used in order to reduce gender information in Experiment 5 and for masking purposes in Experiment 4. Besides the twenty neutral faces, every experiment except Experiments 4, 5, and 7 contained 10 angry, happy, and fearful faces per gender. Experiments 4 and 5 lack the fearful faces while Experiment 7 was conducted without the happy-condition. Thus the number of faces per emotion equalled the number of neutral expressions. Cue size varied between experiments and was either 256 or 512 pixels (for an overview see [Table 1](#)). All cues were presented at the y-axis of the screen with varying eccentricity. Depending on cue size, this implied that faces were presented side by side or with a gap between them. The cue persons displayed at the right and left part of the screen could be of a different or of the same identity. This variation was used in order to test whether we might obtain an effect when emotion is the only discriminating and therefore extremely salient feature. Targets were a circle (15 pixels in diameter) and a square (15×15 pixels) drawn in Psychopy for all experiments except the first one. Here participants reacted on the orientation of two vertically or horizontally arranged dots. Targets were presented exactly at the same positions as the centre of the cue faces.

Procedure

Participants were seated at a distance of 57 cm in front of the screen in separate cabins. After completing the informed consent sheet and personal data, participants were handed a written instruction and the experimenter made sure that the task was properly

Table 1. Overview of experimental conditions in all reported studies.

	Experiment 1	Experiment 2	Experiment 3	Experiment 4	Experiment 5	Experiment 6	Experiment 7
<i>n</i>	39	40	39	60	58	39	33
CTI (in ms)	150	150	100 vs. 300	142 vs. 166	24 vs. 150	500 vs. 1000	150
Cue duration (in ms)	150	150	100	24 vs. 48	24 vs. 150	452 vs. 952	150
Cue size (w × h)	512 × 512	512 × 512	256 × 256	128 × 165	128 × 165	512 × 512	256 × 256
Cue face relation	same	different	different	same	same	different	same
Cue shape	rectangular	rectangular	rectangular	oval	oval	rectangular	rectangular
Emotions	angry, happy, fearful	angry, happy, fearful	angry, happy, fearful	angry, happy	angry, happy	angry, happy, fearful	angry, fearful
#Trials/emotion condition	70	120	480	160	160	320	80
Specifics	additional trials with frequency filtered images	additional trials with frequency filtered images		additional trials with frequency filtered images; masked cue presentation (pre- and postmask, 118 ms each)			additional trials with frequency filtered images

understood. The rest of the experiment was done by the participants alone at the computer. Participants gave their reactions via keyboard in Experiment 6, and via response pad in all other experiments.

Each experiment was preceded by up to 24 additional practice trials. Faces for those trials were also taken from the Radboud Faces Database (Langner, Dotsch, et al., 2010), but did not appear in the experimental trials.

Trials started with the presentation of a white fixation cross on a grey background that remained on the screen until the end of the trial (see Figure 1). Then, two horizontally arranged cue faces, located at 256 pix left and right to the fixation cross appeared. Importantly, in each trial one of the cue faces was neutral while the other one had an emotional expression. The spatial position of the emotional cue was counter-balanced. Cues were followed by a target display. The two simultaneously presented cues could be of the same or a different identity (depending on experiment), but were always of the same gender. Cue positions were fully counter-balanced so that cues had no predictive value for target position. Participants had to ignore the cues and respond to the target, which remained on the screen for 100 ms and was replaced by a screen containing only the fixation cross until participants pressed a key. In the discrimination task, participants indicated whether the target dots were arranged

vertically or horizontally in Experiment 1, or whether the target was a square or a circle in Experiments 2–7 by pressing one of two vertically arranged keys. Participants had to complete different amounts of trials ranging from 160 in Experiment 7 to 1440 in Experiment 3².

Balancing

In all studies the following factors were fully counter-balanced, as far as they varied within study: validity (i.e. correspondence of emotional face presentation-side and target presentation-side), target type (for discrimination tasks), facial emotion, cue gender, and cue-model (i.e. the exact person shown as cue). In Experiments 2, 3, and 6, two different models were shown in each trial. In those experiments, each model was shown equally often as emotional cue (balanced) while the models for the neutral cue were selected randomly (within the given gender) with the limitation that there were always shown two different identities in one screen. CTI was counter-balanced in Experiments 3 and 6, while it was a between subjects factor in Experiments 4 and 5.

Differences between experiments

Timing. The CTI was either held constant (Experiments 1, 2, and 7) or varied between subjects (Experiments 4 and 5) or between trials (Experiments 3 and 7).

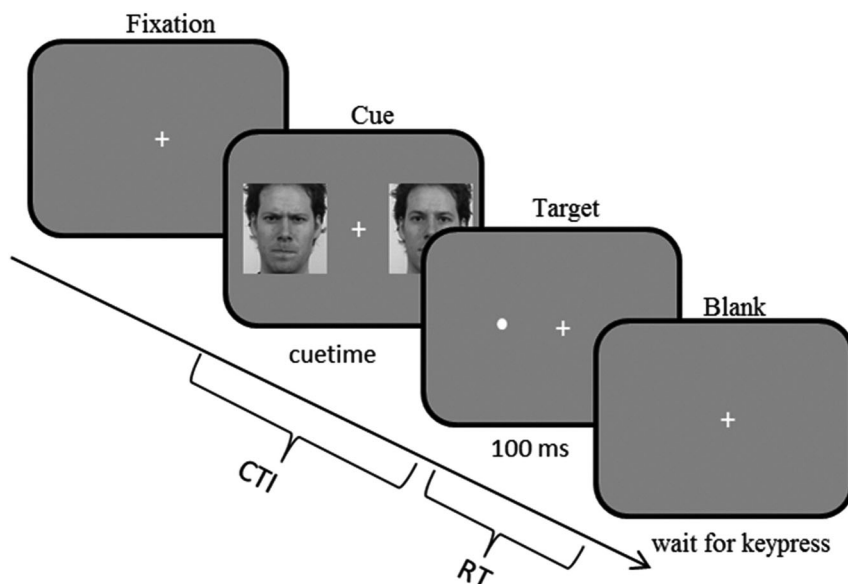


Figure 1. General procedure. Permission obtained.

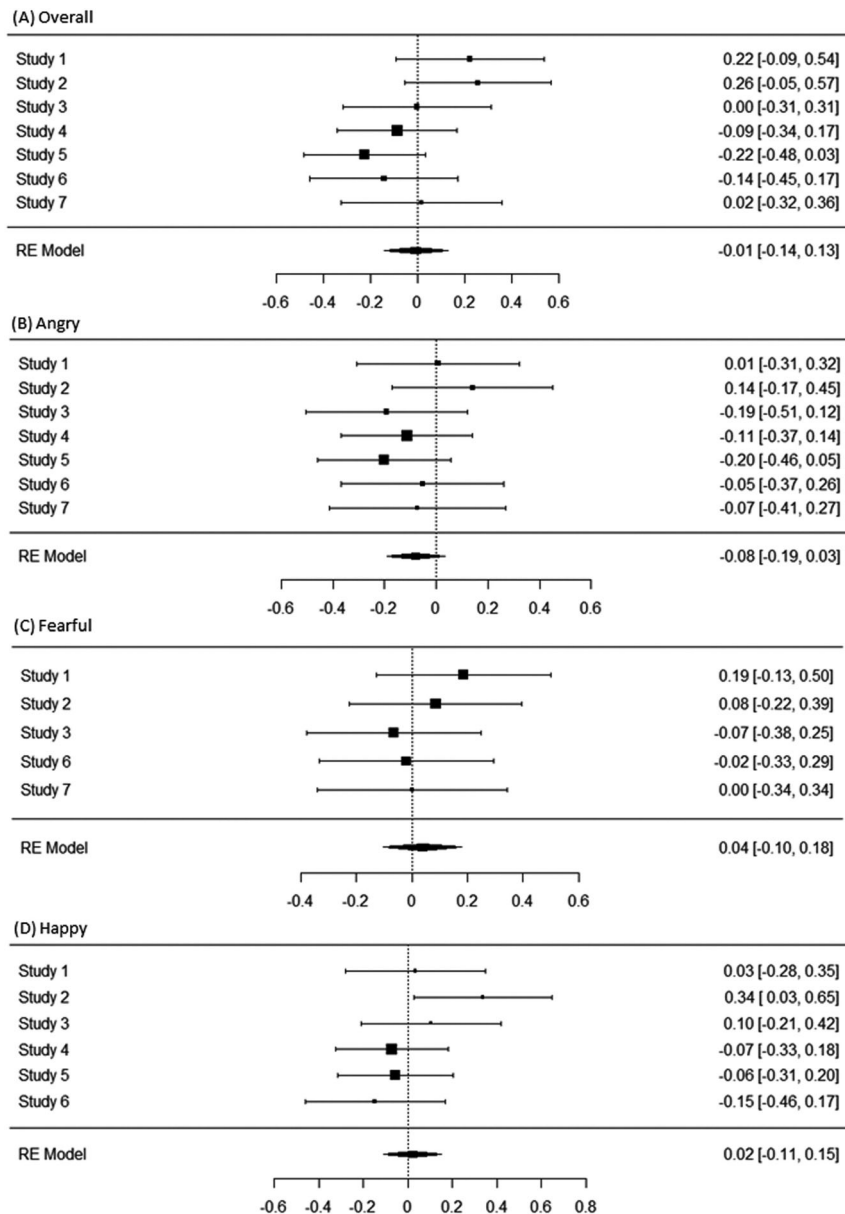


Figure 2. Forest plots on RT validity effects. Each row represents a study with standardised means shown as square and lines representing the confidence interval. Square size indicates sample size. The rhombus in the bottom line shows the emotion validity effect across all included studies.

Additionally, CTI could equal cue presentation time or be higher due to an additional blank screen that was presented after cue offset and target onset. We covered a wide range of CTIs, focusing on short CTIs to guarantee automatic cue processing and to rule out the possibility of inhibition of return effects ($CTI \leq 300$ ms; except for Experiment 6).

Masked presentation. In Experiment 4 we used a black and white oval (same size as faces) with randomly distributed structures that matched faces in their physical characteristics analogously to the masks of Langner, Becker, and Rinck (2010). This masking procedure was implemented to include a condition where emotions of the face cues could not

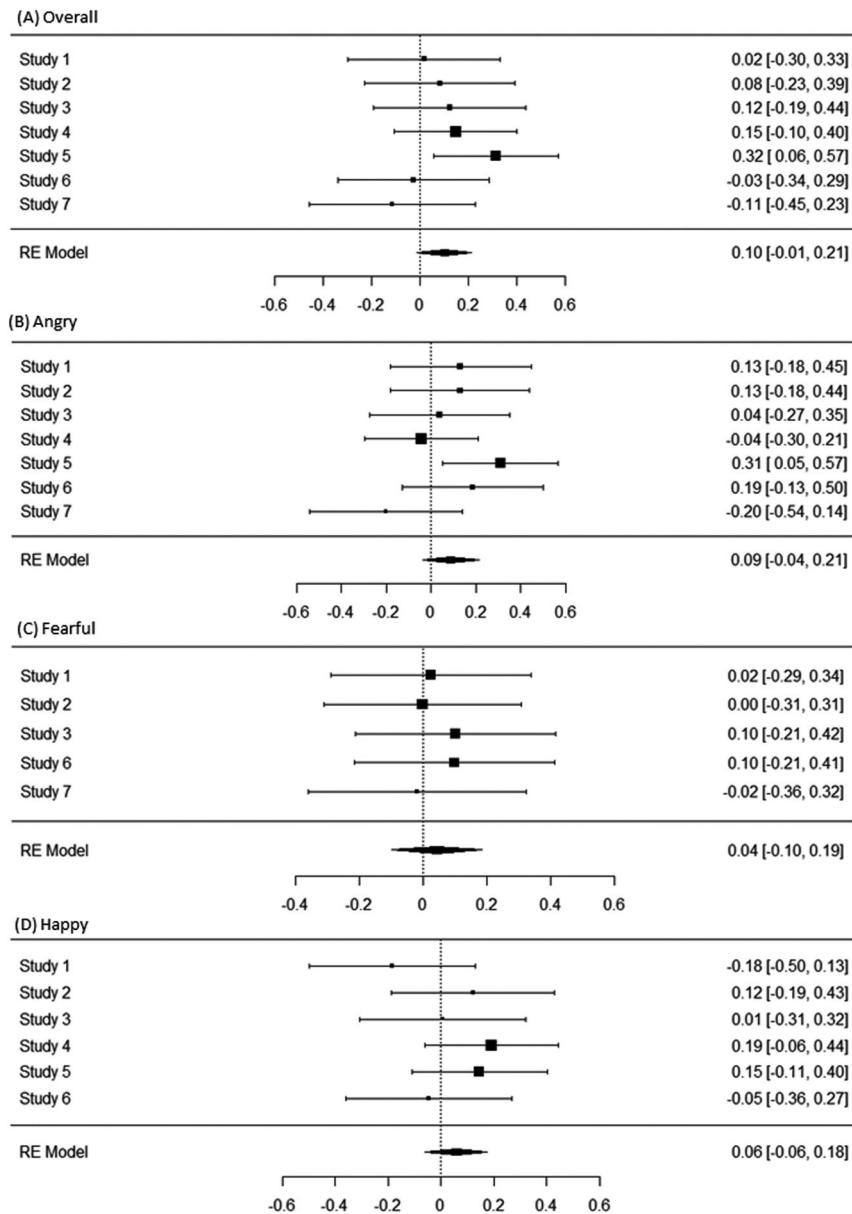


Figure 3. Forest plots on %error validity effects. Each row represents a study with standardised means shown as square and lines representing the confidence interval. Square size indicates sample size. The rhombus in the bottom line shows the emotion validity effect across all included studies.

be identified consciously and thus effects of attentional capture have to occur automatically.

Additional variations. In addition to the aforementioned theoretically important variations, studies differed in other presentation parameters due to the respective original research question. The exact parameters can be found in [Table 1](#).

Results

RT analyses

RT analyses were conducted with correct responses only. Within studies we excluded reaction times that were more than 1.5 interquartile distances above the 75th percentile of the individual distribution separately for each validity condition.

Table 2. Validity effects with means and standard deviations, ANOVA results (F -values) for global validity effects with data aggregated across all experimental conditions, and interactions of validity effects with type of emotion and CTI, calculated separately for each study.

(A) RTs										
Experiment	N	MeanRT	Validity effects in ms (SE)				ANOVA (F -values)			
			Glob	Angry	Fearful	Happy	Glob F	Val $\times$ Emo	Val $\times$ CTI	Val $\times$ Emo $\times$ CTI
1	39	457	3 (2)	4 (3)	4 (4)	1 (4)	1.87 ns	<1 ns	–	–
2	40	464	4 (2)	3 (3)	2 (3)	6 (3)	3.64 ns	<1 ns	–	–
3	39	442	0 (2)	–4 (2)	1 (3)	2 (3)	<1 ns	2.05 ns	<1 ns	1.03 ns
4	60	440	–1 (1)	–2 (2)	–	0 (2)	<1 ns	1.05 ns	1.47 ns	<1 ns
5	58	450	–2 (1)	–2 (2)	–	–1 (2)	1.69 ns	<1 ns	<1 ns	<1 ns
6	39	477	–1 (1)	0 (2)	–1 (1)	–2 (2)	<1 ns	<1 ns	<1 ns	2.50 ns
7	33	474	–1 (3)	–1 (4)	–2 (4)	–	<1 ns	<1 ns	–	–
(B) Error Rates										
Experiment	N	MeanErr	Validity effects in Err (SE) %				ANOVA (F -values)			
			Glob	Angry	Fearful	Happy	Glob F	Val $\times$ Emo	Val $\times$ CTI	Val $\times$ Emo $\times$ CTI
1	39	8.08	0.05 (0.45)	0.85 (0.86)	0.24 (1.06)	–0.95 (0.82)	<1 ns	<1 ns	–	–
2	40	8.08	0.32 (0.61)	0.49 (0.86)	–0.11 (0.87)	0.58 (0.90)	<1 ns	<1 ns	–	–
3	39	9.61	0.28 (0.36)	0.20 (0.63)	0.50 (0.62)	0.13 (0.61)	<1 ns	<1 ns	<1 ns	<1 ns
4	60	4.86	0.32 (0.28)	–0.08 (0.44)	–	0.72 (0.41)	1.31 ns	1.55 ns	<1 ns	<1 ns
5	58	4.69	0.75 (0.31)	1.00 (0.43)	–	0.49 (0.51)	5.81*	<1 ns	<1 ns	<1 ns
6	39	4.16	–0.03 (0.16)	0.26 (0.29)	–0.22 (0.32)	–0.12 (0.33)	<1 ns	<1 ns	1.31 ns	4.21*
7	33	7.36	–0.53 (0.82)	–1.02 (1.03)	–0.04 (1.04)	–	<1 ns	<1 ns	–	–

* $p < 0.05$.

Validity effects

We calculated validity effects separately for every study and cue emotion. Therefore we averaged RT for invalid and valid trials and calculated the difference of invalid – valid trials. Results can be found in Table 2 (for a detailed view on the mean RTs for every emotion by validity condition see Appendix A). None of the experiments showed an attentional capture effect for any of the emotion conditions nor could we find an effect aggregated across all emotions in any of the experiments. There was no moderation of validity effects by CTI or emotion. Additionally we calculated *post hoc* power with the help of GPower (Faul, Erdfelder, Lang, & Buchner, 2007). While Power for a small effect size ($d_z = 0.2$) was insufficient for all studies (median $(1-\beta) = 0.34$, range 0.30–0.45), power to find an effect with medium effect size ($d_z = 0.5$) was high throughout (median $(1-\beta) = 0.93$, range 0.88–0.99), ruling out that effects of this size were overlooked.

Joint analysis

To obtain a higher statistical power, we also conducted an analysis that combined data of all experiments that contained all three emotions, treating Experiment as an additional factor (Experiments 4, 5, and 7 had to be excluded from this analysis, because they contained only two emotional expressions). For this analysis, we could not use CTI as a factor since we did not manipulate CTI in a fully balanced design across experiments. CTI conditions within an experiment were thus aggregated

for this analysis. An ANOVA with Validity and Emotion as within subjects factors, and Experiment as a between subjects factor also yielded no significant results involving the validity factor. The only significant effect was a main effect of Experiment, $F(3,153) = 5.02$, $p > .01$. The theoretically important F -values were $F(1,153) < 1$, for the validity effect, $F(3,153) = 1.45$ (n.s.), for the interaction of validity and study, $F(2,306) < 1$, for the interaction of validity and emotion, and $F(6,306) < 1$, for the three-way interaction, respectively. A power-analysis with GPower (Faul et al., 2007) showed that the power to find a small effect ($d_z = 0.20$) was $(1-\beta) = 0.97$ for this analysis, that is, the probability to miss an actually existing small effect in this analysis was below .05.

Meta-analysis

To account for the different sample sizes of each study, we additionally calculated validity effects with a meta-analytical approach. As in the previous analysis, we aggregated data within experiments across CTIs. These analyses were conducted for global validity effects (aggregating across all emotions), and also for every emotion separately (including those experiments that contained stimuli with the respective emotion). The forest plots (Figures 2 and 3) list all used experiments one below the other. The squares represent mean validity effects with lines showing the confidence interval and square size representing sample size. The rhombus in the

bottom shows the weighted validity effect across all studies. It does not differ significantly from zero in any of the four plots.

Error analyses

The same analyses that were conducted for RT data were repeated for accuracy rates (see Table 2). While there was no significant effect involving validity in six of the seven experiments, we did find a significant global validity effect in Experiment 5, that is, participants made significantly more errors in invalid compared to valid trials in that experiment. Another significant F -value obtained in Experiment 6, $F(2,76) = 4.21$, $p < .05$, for the three-way interaction of validity, emotion, and CTI. Following up on this interaction in Experiment 6, planned contrasts further show a significant two-way interaction of emotion and validity within the 500 ms CTI condition, $F(2,76) = 3.13$, $p < .05$, but not in the 1,000 ms CTI condition, $F(2,76) = 2.06$ (n.s.). In the short CTI condition, there was a validity effect for angry and happy faces, but a reversed tendency for fearful cues. In the meta-analysis for the accuracy data we see that especially in Experiments 4 and 5, the pattern of errors is the opposite to what was found in the RT data. Effects for the error variable might thus have been influenced by speed-accuracy-tradeoffs. Nevertheless, also in error data neither the overall meta-analysis nor the separate meta-analyses for each of the emotions yielded any substantial effect.

Discussion

In the present study, we systematically varied experimental parameters in a dot-probe paradigm to test whether and under which circumstances emotional faces might elicit automatic attention allocation. Participants were presented with two face cues, one of which always was emotional, whereas the other was neutral. In our sample of normal (i.e. non-clinical, non-anxious) students, we did not find any reliable evidence for emotional validity effects, neither for any of the emotions that were used in our study (angry, fearful, and happy), nor for RTs or error rates aggregated across emotional expressions. Furthermore, none of the variations between studies regarding CTI or other presentation parameters seemed to have a systematic influence on the results. *Post hoc* power-analyses revealed that power was sufficient to detect medium effects in every single study ($(1-\beta) \geq .88$), and to detect small effects in the overall-

analysis ($(1-\beta) = 0.97$). The implication of these analyses is that if an effect for either a specific emotion or emotional faces in general existed, it would have to be extremely small. Analyses of accuracy data yielded comparable results. We found null-effects in all experiments except one (Experiment 5), where a small but significant general emotion validity effect appeared, and another rather enigmatic three-way interaction in Experiment 6 (using long CTIs) that was hardly interpretable. Again, meta-analyses revealed that validity effects were not reliable for single emotions and also not for aggregated indicators of automatic attention allocation across emotions (global validity effect).

Various explanations can account for the absence of validity effects in most of the experiments. The simplest one is that people do not attend automatically to emotional faces, at least not in a dot-probe paradigm, in which facial emotions of cues are completely task irrelevant. Such a sweeping conclusion, however, requires a closer look at the experimental parameters and sample characteristics of the present study. As we have mentioned earlier, a reliable validity effect for angry faces has been reported in other studies employing clinically or sub-clinically anxious participants, indicating automatic attentional capture by angry faces for this population (Arndt & Fujiwara, 2012; Bar-Haim et al., 2007; Fox, 2002; Mogg, Bradley, Miles, & Dixon, 2004; Stevens, Rist, & Gerlach, 2009).

Besides sample characteristics, another factor of our studies may seem somewhat atypical for dot-probe research: Because CTI was an important factor that was explored and manipulated in many of our experiments, we chose to hold presentation times of the cues constant, to not confound CTI with cue presentation times. This is why we decided to keep presentation times constant at or below 150 ms (except Experiment 6; for details see Table 1). Compared to other studies, this is a rather short cue presentation time (see meta-analyses of Bantini, Stevens, Gerlach, & Hermann, 2016; Bar-Haim et al., 2007). Thus one could argue that emotions might not have been fully processed yet when cues vanished. Such an explanation is rather improbable, however, because even when the actual stimulus is not present anymore, the after-image typically can be assumed to persist within the iconic memory for sufficiently long durations (Dick, 1971), unless it is masked by a postmask (Liss, 1968; Sperling, 1967). That is, processing of the cues should have proceeded even when they were no longer presented on the screen, with the exception of the masking experiment (Experiment 4).

Another parameter worth considering regards the selection of the emotional face cues. In all our studies, the same faces database was used, drawing on a well-validated set of pictures of emotional facial expressions (Radboud faces database, Langner et al., 2010). In this database, models were intensively trained with a detailed training manual based on the Investigator's Guide for the Facial Action Coding System (Ekman & Friesen, 1978). Afterwards, all emotional expressions were successfully validated in a rating procedure. The agreement of chosen and targeted emotions was 82% (Langner et al., 2010). Importantly, we selected only three emotions for our experiments (angry, happy, and fearful), which are the most important and well-researched emotional expressions in the literature. Of course, however, our findings cannot be generalised to other emotional expressions, that is, we do not know whether disgusted, sad, or surprised expressions might have captured attention. Nevertheless, the selection contains evolutionary important expressions, which can be expected to elicit attentional effects of at least equal strength than the emotions that were not included.

Much more likely appears the assumption that emotional capture effects in the dot-probe paradigm for facial expressions of emotion in normal, non-clinical samples might reflect a publication bias. For instance, in a dot-probe study with words and threatening scenes, Schmukle (2005) reported a low internal consistency as well as a low retest reliability and concluded that the dot-probe paradigm was unreliable. In their meta-analysis, Bar-Haim et al. (2007) did not find dot-probe effects for threatening pictures in non-anxious participants.

What we also should entertain is the possibility that the failure to find attentional capture effects for facial expressions of emotions might not be restricted to the dot-probe paradigm, implying that there is nothing like unconditional automatic attentional capture for these stimuli. Indeed we investigated automatic attentional capture for emotional facial stimuli in another paradigm, namely the flanker task (Eriksen & Eriksen, 1974). In a series of studies, we did not find reliable evidence for stronger interference effects of emotional compared to neutral stimuli (i.e. a stronger slowing when flankers were emotional compared to neutral), nor did we find consistent evidence for differences between emotional and neutral flankers with regard to compatibility effects (i.e. faster reactions in trials where flanker and target require the same response compared to trials in which flanker and

target are incompatible; Puls & Rothermund, 2017). Relating these findings to other failures to find reliable evidence for automatic attentional capture of emotional faces that were reported in the literature (e.g. Barratt & Bundesen, 2012), these results prompt the conclusion that the null findings of the present study may not reflect an unreliability of the dot-probe task. Instead, our findings may also suggest that emotional expressions in faces do not capture attention in an unconditionally automatic fashion. Such a view would mark a fairly strong deviation from what is commonly assumed with regard to automatic attention to expressions of emotion in healthy well-adapted individuals. These strong conclusions, however, clearly require a much more systematic analysis of all existing data, specifically, we need a meta-analysis of published and unpublished data across different paradigms and parameters.

Rather than ending on such a more skeptical note, we want to conclude with another theoretical possibility: Instead of denying the existence of automatic attentional capture by emotions, in our view, attention allocation to emotional expressions should be considered as a *conditionally* automatic phenomenon. That is, in order to be processed effortlessly and spontaneously, emotions have to be relevant for a person in the respective situation. This theoretical stance is inspired by research on task and context dependence in automatic affective processing (e.g. Rothermund, 2011; Spruyt, De Houwer, & Hermans, 2009; Spruyt, De Houwer, Hermans, & Eelen, 2007), and it also resembles contextual approaches to other phenomena of automatic processing in social cognition (e.g. activation of stereotypes and prejudice) that are also now considered to reflect context-dependent, conditionally automatic phenomena rather than global, rigid, and unconditionally automatic biases (Blair, 2002; Casper, Rothermund, & Wentura, 2010, 2011; Müller & Rothermund, 2012; Wittenbrink, Judd, & Park, 2001). Against this background, attentional capture for specific emotional stimuli should be much more likely to occur if these stimuli are made relevant within the situational context of the experiment. This perspective also explains why robust attentional capture effects are typically found for clinical samples for whom specific emotional expressions represent chronic concerns (e.g. angry faces for socially anxious people).

Such a contextual perspective on emotional capture also has implications for the theoretical question that marked the starting point of our

investigation, that is, the role of spatial frequencies. Our current findings suggest that manipulations of frequency bands will not change effects of unconditionally automatic attentional capture for emotional faces in the dot-probe task: Reducing the information that is contained in a stimulus by filtering out certain spatial frequencies will probably not produce effects that were not present for the unfiltered faces. If, however, reliable attentional capture is found for certain emotional faces in social contexts in which processing of emotional facial information is relevant, then introducing spatial filtering may be a very interesting technique that can inform us whether these effects are mediated via cognitive tuning towards specific frequency bands.

Conclusion

With seven dot-probe studies covering a wide range of systematic variations of experimental parameters, we did not find evidence for unconditionally automatic attentional capture effects for emotional facial expressions. To decide whether this lack of validity effects is due to unreliability of the task or to an actual absence of automatic attentional capture, future research should systematically investigate these effects with alternative paradigms, populations, and contextual manipulations.

Notes

1. We thank an anonymous reviewer for pointing out that these seemingly contradictory results can also be explained by assuming a general threat bias that leads to early attentional capture for threatening stimuli, followed by an avoidance response (orienting away from the threat) during later processing stages.
2. In Experiments 1, 2, 4, and 7, we presented additional trials with faces that were filtered for specific spatial frequency bands. This manipulation was part of our original research question but was not relevant for the present purpose. Therefore trials with manipulated spatial frequency information were discarded from all analyses.

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Appendix

A. Table of mean RTs and mean error rates separately for every emotion and validity condition

Experiment	RT in ms						Error in %					
	Angry		Fearful		Happy		Angry		Fearful		Happy	
	valid	invalid	valid	invalid	valid	invalid	valid	invalid	valid	invalid	valid	invalid
Experiment 1	455	460	454	458	456	457	7.41	8.26	8.41	8.66	8.32	7.38
Experiment 2	462	465	465	467	459	465	7.48	7.97	8.72	8.61	7.56	8.15
Experiment 3	444	440	442	444	440	443	9.21	9.41	9.24	9.75	9.95	10.08
Experiment 4	440	438	-	-	440	440	4.71	4.63	-	-	4.69	5.41
Experiment 5	451	449	-	-	451	449	4.14	5.14	-	-	4.50	4.99
Experiment 6	477	476	477	476	478	477	4.26	4.52	4.24	4.01	4.02	3.90
Experiment 7	474	473	475	473	-	-	7.99	6.97	7.26	7.22	-	-

B. Example of cue image. Permission obtained.



C. List of used images (original filenames from Radboud faces database)

	Female	Male
Main Experiment	Rafd090_01_Caucasian_female_neutral_frontal Rafd090_04_Caucasian_female_neutral_frontal Rafd090_08_Caucasian_female_neutral_frontal Rafd090_14_Caucasian_female_neutral_frontal Rafd090_19_Caucasian_female_neutral_frontal Rafd090_22_Caucasian_female_neutral_frontal Rafd090_32_Caucasian_female_neutral_frontal Rafd090_56_Caucasian_female_neutral_frontal Rafd090_58_Caucasian_female_neutral_frontal Rafd090_61_Caucasian_female_neutral_frontal	Rafd090_05_Caucasian_male_neutral_frontal Rafd090_07_Caucasian_male_neutral_frontal Rafd090_15_Caucasian_male_neutral_frontal Rafd090_21_Caucasian_male_neutral_frontal Rafd090_23_Caucasian_male_neutral_frontal Rafd090_25_Caucasian_male_neutral_frontal Rafd090_30_Caucasian_male_neutral_frontal Rafd090_36_Caucasian_male_neutral_frontal Rafd090_46_Caucasian_male_neutral_frontal Rafd090_71_Caucasian_male_neutral_frontal
Practice	Rafd090_02_Caucasian_female_neutral_frontal Rafd090_12_Caucasian_female_neutral_frontal Rafd090_16_Caucasian_female_neutral_frontal	Rafd090_10_Caucasian_male_neutral_frontal Rafd090_20_Caucasian_male_neutral_frontal Rafd090_33_Caucasian_male_neutral_frontal